

SHTS: An IoT-Based Smart Human-Following Shopping Trolley with Automated Billing and Intelligent Navigation

Abstract—Shopping is difficult with traditional shopping trolleys, especially for elderly and physically challenged people, who are required to do manual tasks repeatedly in order to shop. Furthermore, intelligent technologies are becoming more common in stores, and retailers are increasingly looking to incorporate them to improve the in-store shopping experience. This paper proposes a low-cost real-time human following smart shopping trolley system designed and developed using Arduino platform for automated billing operations. The ultrasonic sensor determines the distance between the user and the system, and keeps a following distance at a fixed point, which moves the trolley accordingly. The RFID technology is used for product identification and automatic billing, and a web-based dashboard for monitoring carts, checkout and receipt generation. The proposed system would be beneficial in the sense that it would decrease the amount of physical work required, make shopping more convenient and make the checkout process more easy. Experimental evaluation of the prototype shows that real-time trolley following, product identification using RFID and automatic billing are feasible, and therefore offers a realistic smart shopping solution for the retail environment.

Index Terms—Smart Shopping Trolley, Arduino, Human following, Ultrasonic sensor, Obstacle detection, Online billing, Real time system, Retail automation.

I. INTRODUCTION

With the advancement of the Internet of Things (IoT), intelligent retail systems have emerged, enhancing the convenience and efficiency of the retail environment [1]. But, conventional shopping carts need to be pushed constantly by the user which may be inconvenient for those who are elderly or physically challenged. The traditional checkout counter also has several drawbacks, such as long waiting times and the need for more human resources, which is why intelligent trolleys are needed to save the human efforts for checkout and simplify the billing process [2].

IoT-based smart shopping trolleys integrate sensors, micro-controllers, and wireless communication to support real-time interaction between customers, carts, and retail systems. These technologies can assist trolley movement, identify products, automate billing, reduce checkout congestion, and improve inventory management. Specifically, ultrasonic sensing can facilitate autonomous human following, and RFID can facilitate automatic product identification and billing [3], [4]. Human following, obstacle detection, RFID billing, and IoT communication have been studied for smart trolleys in recent years [5], [6], [7]. Many of the systems are either automated billing or autonomous motion, but some systems need extra hardware,

extra line-following routines or some existing infrastructure. This paper aims to overcome these drawbacks by proposing a smart shopping trolley based on autonomous human following, RFID automated billing, and intelligent web-based monitoring, which is also low-cost. The major contributions of this work are as follows:

- 1) Design and development of an autonomous human following smart shopping trolley with low cost ultrasonic-based system.
- 2) Implementation of an RFID-based automated billing system for real-time product identification and faster checkout.
- 3) Design and experimental validation of a unified intelligent retail solution that combines autonomous mobility, automated billing, and cost-effective hardware, making it suitable for practical deployment in supermarkets and shopping centers.

II. BACKGROUND AND RELATED WORK

In recent years, several studies have been carried out on smart shopping trolleys and shopping carts to enhance the efficiency, convenience and automation of the shopping process in supermarket stores. The majority of the current systems are designed to automate billing, identify products, provide customer support services, guide navigation and real-time monitoring via technologies like RFID, IoT, AI, and wireless communication. These systems are primarily designed to streamline the checkout process, save time and labor, and enhance the overall shopping experience.

In [8] a smart shopping cart system using RFID and QR code technology was presented, which can automatically recognize the products and streamline the billing process. Hyun et al. [9] emphasized the system efficiency, usability and user participation in e-shopping which can be an inspiration for designing a user friendly smart shopping system. Mamatha et al. designed a hybrid RFID/QR smart cart that automatically bills and monitors the cart, but their extra infrastructure makes their system more complex and expensive [13].

Padmavathi et al. [14] introduced an IoT based human following smart shopping trolley for RFID based billing. The system is a mixture of trolley movement and automatic billing that enables the customer to move without pushing the trolley. This results in a more convenient approach but multiple hardware and communication components may introduce complexity into the system.

TABLE I: Comparison of Existing Smart Shopping Trolley Systems and the Proposed System

Ref.	Human Follow.	Auto. Billing	RFID	Web Billing	Low Cost
Hyun et al. [9]	×	✓	×	×	✓
Mamatha et al. [13]	×	✓	✓	×	×
Chen et al. [19]	×	×	×	×	×
Eirale et al. [17]	✓	×	×	×	×
Jothi et al. [15]	×	✓	✓	×	✓
Padmavathi et al. [14]	✓	✓	✓	×	×
Matthew et al. [18]	×	✓	✓	×	✓
Zulfiqar et al. [16]	✓	✓	✓	×	×
Maurya et al. [20]	×	✓	✓	×	×
Jain et al. [21]	×	✓	✓	×	✓
Proposed	✓	✓	✓	✓	✓

Jothi et al. [15] introduced an RFID-based smart shopping trolley drawing on the concept of the Internet of Things (IoT) that automatically bills the customer and tracks cost in real time. The system enables the customer to check product data and cumulative cost as he/she is shopping. It makes checkout more efficient and quicker, but it does not emphasize the human-following aspect.

Zulfiqar et al. [16] proposed an AI-based smart shopping cart for real-time tracking and navigation, while Eirale et al. [17] surveyed human detection, tracking, motion planning, and navigation methods for autonomous robots. Matthew et al. [18] developed an RFID-based smart cart for automatic product recognition and instant billing, reducing checkout time, but without human-following or autonomous movement.

The literature review reveals that most of the smart shopping trolleys are focused on the one of two aspects: automated billing or movement intelligence, but very few look at both aspects. Many need expensive hardware or infrastructure or high computing power as well. Table I is a comparison of the systems reviewed.

III. METHODOLOGY AND SYSTEM DESIGN

The proposed methodology involves the overall design and operating flow of the Arduino based Smart Human following shopping trolley. The system combines the human following functions based on the HC-SR04 ultrasonic sensor and the product identification and the automatic billing functions based on RFID. The Arduino Uno is the control unit that takes in the sensor readings, controls the trolley's motion and communicates with the RFID reader. Cart monitoring, bill generation, checkout and receipt generation are all accomplished using a web based dashboard.

A. System Architecture

The overall architecture of the proposed smart shopping trolley is shown in Fig. 3. The system consists of an Arduino Uno, an HC-SR04 ultrasonic sensor, an RFID reader, an L298N motor driver, and DC motors. The Arduino Uno acts as the central controller by processing sensor data and controlling the trolley movement. The ultrasonic sensor enables human following and obstacle detection, while the RFID reader identifies products for automated billing. The web dashboard

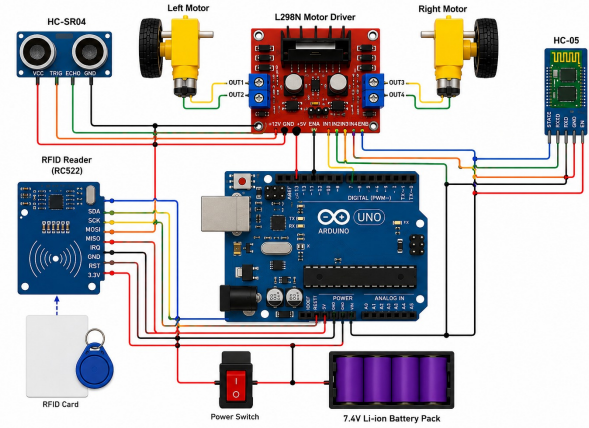


Fig. 1: Circuit diagram of the proposed smart shopping trolley.

processes billing in real time, allowing users to monitor and manage their shopping bill instantly.

B. Circuit Diagram

The circuit diagram of the proposed smart shopping trolley is shown in Fig. 1. The Arduino UNO is used as the central controller to coordinate the sensing, product identification, and motor control operations. The HC-SR04 ultrasonic sensor is interfaced with the Arduino to measure the distance between the user and the trolley for implementing the human-following mechanism. The RC522 RFID reader is connected to the Arduino to identify RFID-tagged products during the shopping process. The L298N motor driver receives control signals from the Arduino and drives the four DC motors responsible for the movement of the trolley. A 7.4 V Li-ion battery pack is used as the primary power source, with a power switch provided for system activation and deactivation. The circuit therefore integrates distance sensing, RFID-based product identification, motor actuation, and power supply into a single hardware platform for the proposed smart shopping trolley.

C. System Workflow

The flowchart of the proposed smart shopping trolley is shown in Fig. 2. Arduino Uno, sensors, and the billing system are initialized initially. The web dashboard continuously updates the shopping bill in real time as RFID-tagged products are scanned. Once the trolley is initialized, it goes into human-following mode. The survey of the Ultrasonic sensors is continuously carried out and the movement of the trolley is calculated. The trolley moves forward, left, right and stops based on the readings from the various sensors. This will be repeated again and again throughout the shopping session until it is over.

D. Algorithm of the Proposed System

Algorithm 1 describes the operational sequence of the proposed smart shopping trolley. It has a system initialization, customer tracking, obstacle detection, billing with RFID during the shopping period.

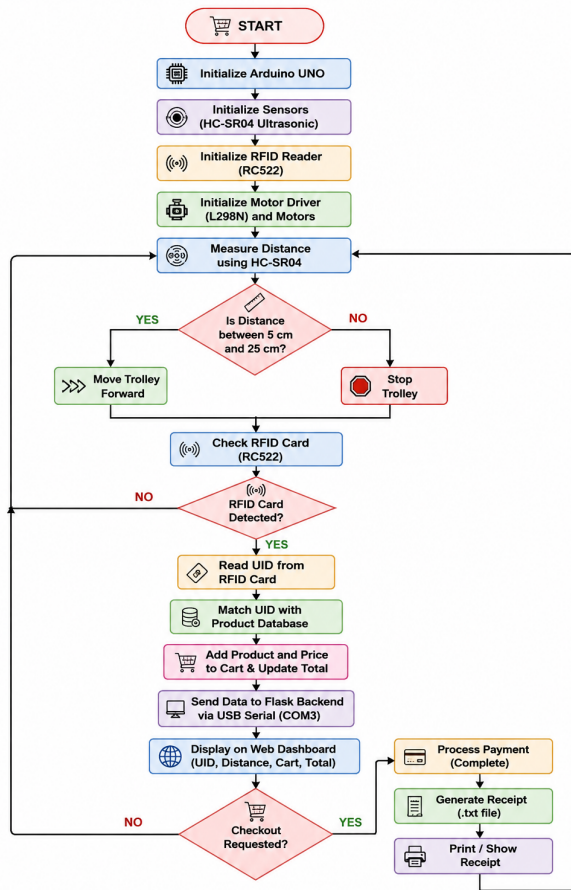


Fig. 2: Operational workflow of the proposed smart shopping trolley.

E. Web-Based Billing Process

The web-based billing process offers an easy interface for product management, cart items, checkout and billing. Once the system is up and running, the product catalogue is loaded and contains product names and prices. Users are able to add products to the cart, change quantity and see the new subtotal and running total.

At check-out, the system checks whether the cart is empty or not, and sums the prices and quantities of the products. Once confirmed, the transaction is complete, a receipt is generated and the cart is emptied for the next shopping session.

There is also an administration module to manage the product catalogue. Admin can add and remove products and change prices. Simple checks are done to ensure that there is no invalid product information and no empty and incorrect transactions, which in turn increases the reliability of the web-based billing system.

Table II summarizes the major hardware components used in the proposed smart shopping trolley and their respective purposes. Each component is selected to support a specific system functionality, including human following, product identification, motor control, and power management. The integration of

Algorithm 1 Smart Human-Following and Automated Billing Algorithm

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1: Initialize Arduino Uno, HC-SR04, RFID Reader, and
   Motor Driver
2:  $TB \leftarrow 0$  ▷ Initialize total bill
3: while System is ON do
4:   Measure distance  $d$  using HC-SR04
5:   if  $5 < d < 100$  then
6:     Move trolley forward
7:   else
8:     Stop trolley
9:   end if
10:  if New RFID card is detected then
11:    Read RFID UID
12:    if UID matches a registered product then
13:      Identify product and retrieve price  $P_i$ 
14:       $TB \leftarrow TB + P_i$ 
15:      Send product and UID data to Flask via USB
        Serial
16:    else
17:      Display "Unknown Card"
18:    end if
19:  end if
20: end while
21: Display total bill  $TB$ 

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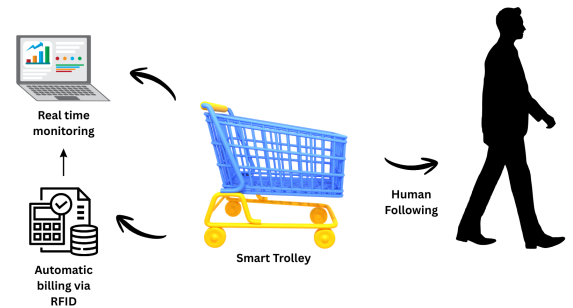


Fig. 3: System architecture of the proposed smart shopping trolley.

these components enables the trolley to perform autonomous movement and RFID-based automated billing efficiently.

Table III presents the hardware cost breakdown of the proposed smart shopping trolley. The estimated cost includes the Arduino UNO, chassis kit, motor driver, ultrasonic sensor, RFID reader, and RFID cards required for prototype development. The total hardware cost is \$32.00, demonstrating that the proposed system can be implemented as a relatively low-cost smart shopping solution.

TABLE II: Hardware Components and Their Purpose

Hardware	Purpose
Arduino Uno	Controls the entire system and processes sensor and RFID data.
Chassis Kit	Provides the mechanical structure and holds the system components.
L298N Motor Driver	Controls and drives the DC motors of the trolley.
Ultrasonic Sensor (HC-SR04)	Measures the distance between the trolley and the user for human-following operation.
RFID Reader (RC522)	Reads RFID card UID for product identification and automated billing.
RFID Card	Stores the unique UID associated with each registered product.

TABLE III: Hardware Cost Breakdown of the Proposed Smart Shopping Trolley

Component	Unit Cost (\$)	Quantity	Total (\$)
Arduino Uno	10.50	1	10.50
Chassis Kit	12.00	1	12.00
L298N Motor Driver	3.50	1	3.50
Ultrasonic Sensor (HC-SR04)	2.00	1	2.00
RFID Reader (RC522)	3.00	1	3.00
RFID Card	0.50	3	1.50
Total			32.00

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The proposed Smart Human-Following Shopping Trolley was implemented and tested experimentally in order to determine its human following capability, its automated RFID-based billing feature and its web-based bill monitoring feature, as well as the overall system performance. The designed prototype of Arduino was used for the experiments under various operating condition. The human following performance was judged by observing the movement of the trolley at varying distances between the user and the trolley. The RFID billing system has been tested with the registered product cards for the identification of the product, calculation of price, updating of the cart and generation of the total bill. Also, the web dashboard (Flask) was tested to show real-time RFID data, distance, items on the cart and total bill, as well as checkout and receipt production. The results obtained are discussed in the subsections below.

A. Prototype Implementation

An Arduino Uno was used as the central controller for the proposed Smart Human-Following Shopping Trolley and it was implemented as a low-cost prototype. The prototype consists of HC-SR04 ultrasonic sensor, which acts as a distance sensor between the trolley and user, RC522 RFID reader that reads the product identification tag, L298N motor driver to drive DC motors and trolley chassis for mechanical movement. The Arduino receives the information of the sensor and according to this information it moves the trolley. Every

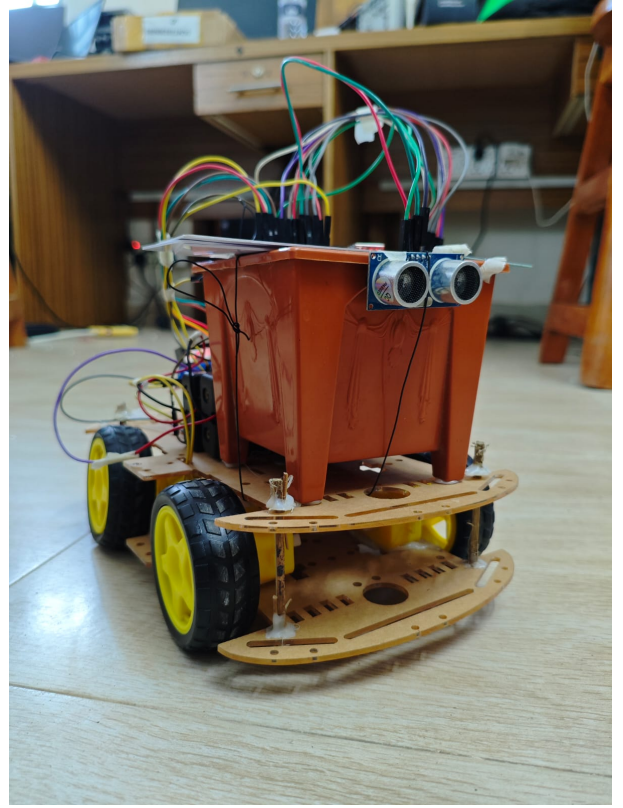


Fig. 4: Prototype implementation of the proposed Smart Human-Following Shopping Trolley.

registered product has an individual RFID card that enables identification of the product and then it calculates the bill value. The developed prototype is shown in Fig. 4.

B. Human-Following Performance

Experimental testing of trolley for human-following is proposed. The distance to the target in front of the person being operated on is measured and the person is operated on. Controlling a trolley with an HC-SR04 ultrasonic sensor. The sensor determines distance is measured using the time for an ultrasonic sensor. The target is able to move to the sensor and back. The distance is calculated as:

$$d = \frac{v \times t}{2} \quad (1)$$

where d is the measured distance, v is the speed of sound in air, and t is the round-trip travel time of the ultrasonic pulse.

The trolleys were programmed to travel in the implemented system forward when the distance measured was within the predefined. It was only able to go from 5 to 100 cm, otherwise it stopped. Experimental distances measured were 2.56, 4.06, 7.32, 12.88, and 21.71 cm, as well as measurements above 100 cm. The results in Table IV demonstrate that the sensor could be used for controlling distance. The movement of the trolley over the given range.

The difference in the distance measured in the experimental test is shown in Fig. 5. The graph shows the Distance taken

TABLE IV: Representative Ultrasonic Distance Measurements

Test Condition	Distance (cm)	Expected Action
Very Close Range	2.56	Stop
Minimum Range	4.06	Stop
Following Range	7.32	Forward
Following Range	21.71	Forward
Following Range	47.42	Forward
Following Range	53.30	Forward
Following Range	75.00	Forward
Maximum Range	100.00	Forward
Far Range	120.00	Stop
Far Range	400.00	Stop

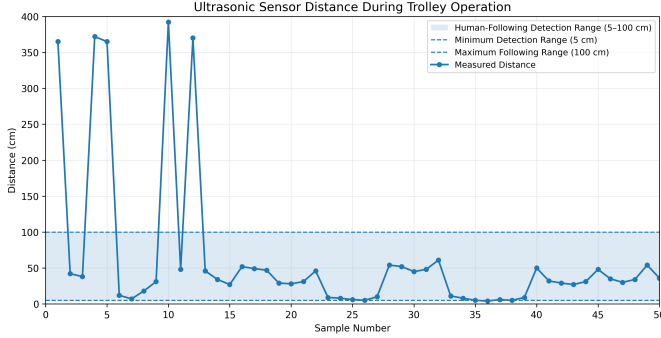


Fig. 5: Ultrasonic sensor distance measurements during human-following operation.

TABLE V: RFID-Based Product Identification and Billing Results

Test	Product	Price (Tk)	Running Total (Tk)
1	Bread	60	60
2	Milk	80	140
3	Biscuit	50	190

by HC-SR04 sensor at various stages of trolley operation. If distance measured within the range of tolerance. The trolley is driven forward if its distance is in the range of 5–100 cm. Remains block is stopped based on the logic of control implemented.

The test distance calculations show that the HC-SR04 The distance information of the sensor can be used as continuous information to control the . trolley movement. The range of measurement observed was 5-25cm. provide for the intended human following operation, measurements If it is outside this range, then the stopping condition is met.

C. RFID-Based Automated Billing

Three registered RFID cards representing Bread, Milk, and Biscuit were used to test the automated billing system. The RFID reader successfully read the UID of each card and identified the product. During testing, Bread, Milk, and Biscuit were assigned prices of Tk 60, Tk 80, and Tk 50, respectively. The running total increased automatically after each RFID scan, resulting in a total of Tk 190, as shown in Table V.

Product information and the running total were also shown when the product is scanned On the Flask based web dash-

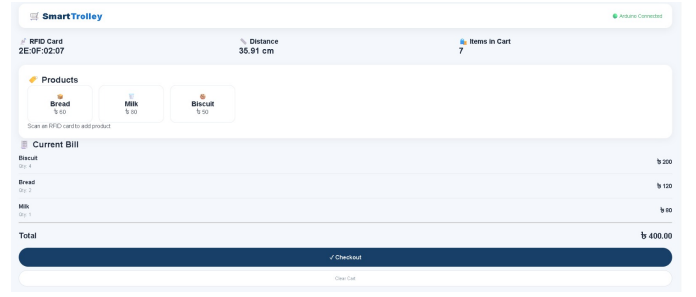


Fig. 6: Web dashboard showing RFID-based product identification and real-time billing.

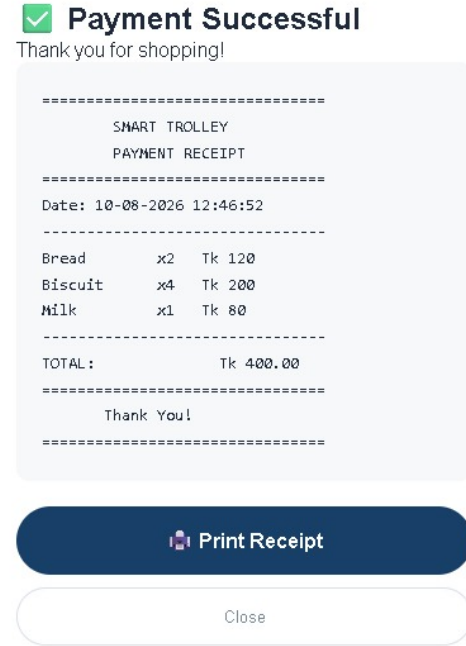


Fig. 7: Generated receipt after completing the checkout process.

board, as depicted in Fig. 6. The dashboard test showed product identification done automatically on the dashboard, Tracking quantities, calculating subtotal and updating the total bill. In the shown test case, there are two Bread items, four Biscuit items, One Milk and one item of Milk costed Tk 400 in total.

After checkout, the system printed out a digital receipt with the The product name, quantity, individual price and total price, shown in Fig. 7. This demonstrates the integration of Product identification, real time billing, web based shopping cart, monitoring and receipt generating aspects.

The results show that the proposed system is able to successfully couples RFID-Identification of products with real-time billing, Automated receipts, web-based cart monitoring, and more.

V. DISCUSSION

The human following mechanism is based on HC-SR04 ultrasonic sensor measuring the distance between the trolley and the human. The trolley will be moving forward if the distance measured is between 5 cm and 100 cm and it will stop if it is not. The system is inexpensive and straightforward, but is subject to environmental factors, multiple users and objects nearby.

The RC522 RFID reader reads the UID of registered products and transmits them to the Flask backend using its USB serial port. The backend uses the UID to update the cart, to calculate the total bill and to display the information on the web dashboard.

Arduino is responsible for distance measurement, reading RFID and controlling motors, while the Flask backend controls product mapping, cart management, billing, updates to the dashboard, etc. The payment/receipt interface is offered as an informational example and not for the purpose of conducting any actual financial transactions.

This prototype is a successful demonstration of human following system, automated billing system using RFID and a web-based bill monitoring system. But it is based on ultrasonic sensing and pre-established RFID-product mapping. Future enhancements could involve adding more obstacles for the robot to avoid, expanding product knowledge, enabling secure online payment integration and implementing computer vision tracking

VI. CONCLUSIONS AND FUTURE WORK

This paper introduced a low-cost smart shopping trolley based on the combination of real-time tracking of human being and the use of RFID for product identification, automatic bill generation, and a web-based monitoring system in a single platform with an Arduino microcontroller. The feasibility of the proposed following distance, identification of products using RFID and automatic checkout and receipt generation was confirmed through experiments. The proposed system integrates these functions, which can help minimize repetitive manual tasks and enhance the overall shopping experience, especially for older adults and those with physical limitations. There are still several aspects of the work to be improved, such as obstacle avoidance, user identification, following accuracy, system robustness in crowded settings, and further integration of intelligent navigation and mobile-based services, all of which will be the focus of future work.

REFERENCES

- [1] L. Kommavarapu, T. Mangu, S. P. Macherla, and B. M. M. Tripathi, "Smart Trolley for Quick Shopping," in *Proc. IEEE Int. Conf. Computing, Power and Communication Technologies (IC2PCT)*, Greater Noida, India, 2024, pp. 141–144, doi: 10.1109/IC2PCT60090.2024.10486478.
- [2] M. Z. Anwar, H. Kibriya, A. Tahir, S. T. Siddiqui, M. Waqas Tahir, and F. Amar, "Smart Trolley with Automated Billing System," in *Proc. 2025 Int. Conf. Emerging Technologies in Electronics, Computing, and Communication (ICETECC)*, Jamshoro, Pakistan, 2025, pp. 1–6, doi: 10.1109/ICETECC65365.2025.11070307.
- [3] M. S. H. M. A. K. S. U. S. and Y. D., "RFID based Smart Trolley Billing System using IoT," in *Proc. 2025 Int. Conf. Intelligent Computing and Control Systems (ICICCS)*, Erode, India, 2025, pp. 141–147, doi: 10.1109/ICICCS65191.2025.10984823.
- [4] A. C. P. S. D. V. S. K. N. S. and S. R., "RFID based Human Following Smart Trolley with Real-Time Pricing," in *Proc. 2025 5th Int. Conf. Pervasive Computing and Social Networking (ICPCSN)*, Salem, India, 2025, pp. 1902–1907, doi: 10.1109/ICPCSN65854.2025.11034910.
- [5] H. Deng, H. Tan, Z. Lin, M. Huang, H. Han, and X. Chen, "Design of Intelligent Shopping Cart Control System," in *Proc. 2025 Int. Conf. Computing, Robotics and System Sciences (ICRSS)*, Sanya, China, 2025, pp. 150–155, doi: 10.1109/ICRSS67786.2025.00037.
- [6] A. Joshi, T. Thuse, P. Thorve, R. Thakur, A. Thote, and I. Thakur, "Hy-Tech Shopping Cart Using Object Detection and Human Following Technology," in *Proc. 2024 IEEE 4th Int. Conf. ICT in Business Industry & Government (ICTBIG)*, Indore, India, 2024, pp. 1–5, doi: 10.1109/ICTBIG64922.2024.10911594.
- [7] S. Modi, M. Modi, V. Alone, A. Mohite, V. K. Borate, and Y. K. Mali, "Smart Shopping Trolley Using Arduino UNO," in *Proc. 2024 15th Int. Conf. Computing, Communication and Networking Technologies (ICCNT)*, Kamand, India, 2024, pp. 1–6, doi: 10.1109/ICCNT61001.2024.10725524.
- [8] M. S., S. K. Badiger, S. K. S., S. J. T. S., and V. V., "RFID and QR Code-Based Smart Shopping Cart System," in *Proc. 6th Int. Conf. Advances in Electrical, Computing, Communications and Sustainable Technologies (ICAECT)*, Bhilai, India, 2026, pp. 1–5, doi: 10.1109/ICAECT68478.2026.11425024.
- [9] M. Y. Hyun, M. L. Teklehaimanot, X. Ju, H. C. Kim, R. Bakrim, and C. P. Y. Mbadouma, "Effects of competence support–frustration misfit in e-shopping cart systems on task engagement and payment behavior," *Journal of Strategic Marketing*, 2026.
- [10] R. M. Bommi, R. Pichamuthu, B. Sundarambal, S. Subramanian, and K. Chockaia, "Implementation of a Vision AI-Based Smart Checkout System Using Edge Processing and Real-Time Product Recognition," in *Proc. Int. Conf. Electronics and Renewable Systems (ICEARS)*, 2026.
- [11] "AI-Driven Multi-Agent Shopping System through E-Commerce System," *International Journal of Computer Trends in Engineering & Computing (IJCTEC)*, vol. 9, no. 2, pp. 463–471, 2026.
- [12] "International Payments via Smart Contracts," *International Journal of Advanced Research in Computer Science and Technology (IJARCT)*, vol. 9, no. 2, pp. 413–416, 2026.
- [13] Mamatha et al., "Hybrid QR and RFID-Based Smart Shopping Cart with Automated Billing and Monitoring," 2026.
- [14] Padmavathi et al., "IoT-Based Human Following Smart Shopping Trolley with RFID Billing," 2025.
- [15] E. Jothi, R. Thenmozhi, and S. Vidhyalaxmi, "IoT-Enabled Smart Shopping Trolley with Automated Billing and Real-Time Cost Tracking," *International Journal of Advanced Trends in Engineering and Management*, 2025.
- [16] Zulfiqar et al., "AI-Based Smart Shopping Cart with Real-Time Tracking and Navigation," 2025.
- [17] A. Eirale, M. Martini, and M. Chiaberge, "Human Following and Guidance by Autonomous Mobile Robots: A Comprehensive Review," *IEEE Access*, vol. 13, 2025.
- [18] A. Matthew et al., "RFID-Powered Smart Cart: Hassle-Free Shopping and Instant Billing," in *Proc. IEEE Int. Conf. Intelligent Innovations in Engineering and Technology (ICINVENTS)*, 2025.
- [19] H. Chen et al., "Human-in-the-Loop Robot Learning for Smart Manufacturing: A Human-Centric Perspective," *IEEE Transactions on Automation Science and Engineering*, vol. 22, 2025.
- [20] S. Maurya et al., "The IoT-Based Smart Shopping Trolley System," in *Proc. IEEE Int. Conf. on Innovative Computing and Advanced Technologies (ICICAT)*, 2023.
- [21] A. Jain et al., "Secure and Smart Trolley Shopping System Based on IoT Module," in *Proc. IEEE Int. Conf. on Contemporary Computing and Informatics (IC3I)*, 2022.